NETRAX

AI safety monitor for PPE compliance, smoke and fire detection

BPUT Hackathon 2026 · Problem Statement 6 · Progress report: work done and outputs

20 / 22	~42 ms	98-175 ms	51 / 52
plan features done or done for now	inference per frame, 3 models (GTX 1650)	edge to dashboard (live)	self-tests pass (1 = open smoke issue)

Report date: 12 October 2026. Every number in this report was measured on our own prototype unless it is marked **assumption**. Parts that are planned but not built are listed as such. Faces in all images are pixelated.

1. Summary

NETRAX watches CCTV, webcam, phone or recorded video **on a local PC (edge)**. For every person it checks each required PPE item for the zone they stand in (helmet, vest, gloves). It detects fire and smoke, and turns each event into one located, prioritised alert. The alert carries zone, camera, worker ID, missing items, severity, confidence and a snapshot.

It works **autonomously**: the laptop speaker calls the worker by their registered ID until they comply, and a siren plays for fire. The supervisor dashboard keeps a human in the loop (acknowledge / escalate / dismiss, reports, analytics). Only alerts and thumbnails leave the edge; raw video never does.

What exists today

Area	What works now
Detection	Person (COCO YOLOv8n) + PPE model (YOLOv8s: helmet, vest, gloves) + fire/smoke model (YOLOv8n), FP32 on GPU
Per-worker logic	Each item judged per person: worn / missing / can't tell; zone rules; 7-of-10 frames + grace; one alert per violation; re-alert on a real repeat
Worker identity	Registration with photos on the dashboard -> W-001 IDs; face match keeps one ID per person; visitors get V-001 (memory only)
Fire / smoke	Colour, white-hot core, flicker, growth, on-person, skin/hair/glare checks; siren + spoken evacuation call
Autonomous response	Windows voice: 'Attention worker W 0 0 1 ... please wear your helmet now', repeats every 20 s; siren for fire
Cameras	Laptop webcam (Welding bay) + mobile phone over Wi-Fi (Warehouse, picture rotated upright); N cameras supported; load manager
Dashboard	Live tiles, alert feed, zones, workers panel, heatmap, shift summary, CSV/PDF reports, ROI calculator
Robustness	Offline alert buffer, camera health ONLINE/DEGRADED/OFFLINE, fast fail for unreachable cameras
Privacy	Faces pixelated in stored snapshots; 30-day snapshot retention; consent-based registration

Data so far: 194 PPE alerts, 42 fire and 39 smoke alerts from our own live tests (22 Sep - 10 Oct 2026); 6 registered worker.

2. Status against the project plan (F01-F22)

ID	Feature (plan section 3)	Status	Evidence / note
F01	PPE detection (helmet, vest, gloves, boots)	**Done**	Helmet/vest/gloves live; safety shoes paused until our own shoe data
F02	Missing item per person + 'can't tell' state	**Done**	Only 'missing' can alert; feet out of view = can't tell
F03	Fire and smoke detection	**Done**	Match, lighter and fire video confirmed live
F04	Multi-object tracking	**Done**	ByteTrack + face ID for one stable ID per person
F05	N-of-M, grace, dedup, cooldown	**Done**	7/10 frames + 1.5 s; re-arm after 5 s for real repeats
F06	Zone rules engine	**Done**	zones.yaml: required PPE + risk per zone
F07	Severity engine	**Done**	Critical / High / Medium / Low by event + zone risk
F08	Multi-stream + health	**Partial**	Webcam + phone live; 4-stream soak measured; recorded demo clips still to record
F09	Backend + DB + snapshots	**Done**	FastAPI + SQLite
F10	Supervisor dashboard	**Done**	Ack / escalate / dismiss, live tiles, feed
F11	Confidence display	**Done**	On boxes and alert cards
F12	Edge optimisation + measured FPS	**Done**	Benchmark table (PyTorch CUDA vs ONNX CPU), 4-stream table
F13	Fire verification + look-alike suppression	**Done**	Plus skin/hair/glare and flame-core rules
F14	Analytics + heatmap	**Done**	7-day chart, zone x hour heatmap, trend insights
F15	CSV / PDF reports	**Done**	Per shift / zone; PDF includes the shift summary
F16	Escalation + notifications	**Done**	Auto-escalation; Telegram if configured, otherwise MOCK
F17	Offline alert buffer	**Done**	SQLite outbox, sync on reconnect
F18	Failure-case gallery + hard-test metrics	**Partial**	Gallery v1 (7 real cases); metrics need our own labelled data
F19	Worker ID	**Done**	Photo registration (face); badge/RFID planned in hardware phase
F20	LLM / summary layer	**Done (Plan B)**	Templated shift summary (no LLM, works offline)
F21	Cost / ROI	**Done**	docs/cost_roi.md + calculator page
F22	Privacy: head blur	**Done**	Faces pixelated in stored snapshots + retention

Also built beyond the feature list: load-aware processing (plan 7.5), snapshot retention (plan 14), autonomous speaker + siren, phone camera support with rotation.

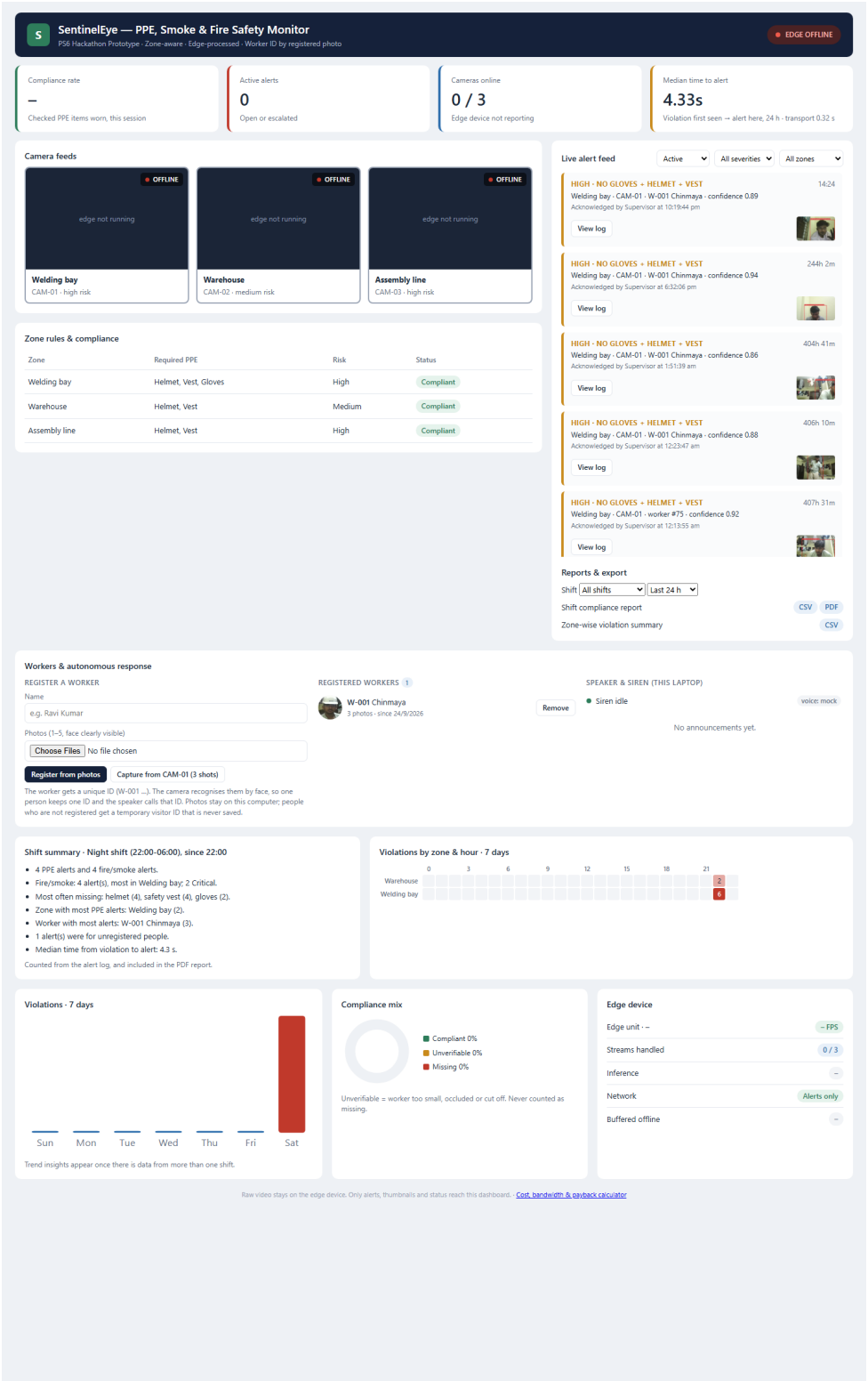
3. Architecture

Everything that touches video stays on the edge PC. Only small alert records (about 42 KB including a snapshot, measured) and status cross to the backend and dashboard.

Layer	Component	File
Sources	Webcam, phone (IP Webcam), RTSP, video files; one reader thread per camera, newest frame only	edge/stream.py
Edge: detection	COCO person + PPE + fire/smoke YOLO models, FP32 on GPU, CUDA warm-up	edge/detector.py
Edge: people	ByteTrack IDs + face identity (YuNet + SFace) -> W-001 / V-001	edge/tracker.py, edge/identity.py
Edge: PPE logic	Item-to-person association, size + hi-vis checks, zone rules	edge/association.py, validate.py
Edge: alert logic	N-of-M, grace, cooldown, re-arm, resolve, ID hand-over	edge/state_machine.py
Edge: fire logic	Colour, white-hot core, flicker, growth, look-alike suppression	edge/fire_verify.py
Edge: system	Severity, load manager, privacy blur, retention, offline outbox	edge/severity.py, load.py, outbox.py
Backend	Events, actions, escalation, reports, analytics, workers, WebSocket	backend/main.py, reports.py
Response	Windows voice (worker ID) + siren	backend/actuators.py
Dashboard	Single-page supervisor UI + ROI calculator	dashboard/index.html, roi.html

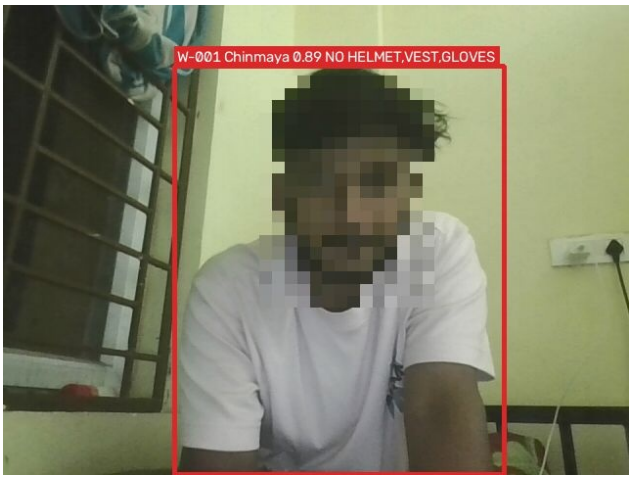
4. Outputs

4.1 Supervisor dashboard (real data from our tests)

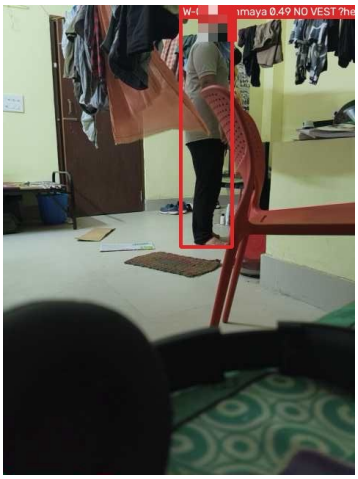


Screenshot taken from the running backend; cameras are shown offline because the edge was stopped for the screenshot. Alert feed, workers panel, shift summary and heatmap use the real event log.

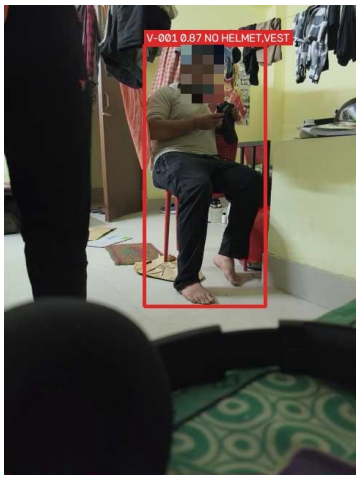
4.2 Live detections from our tests (faces pixelated)



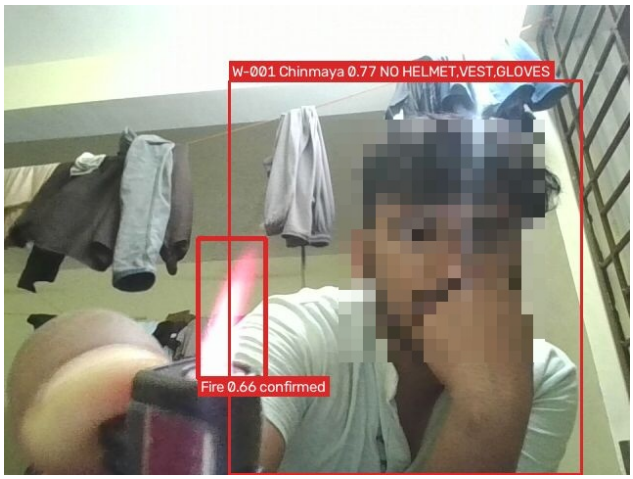
10 Oct - Webcam, Welding bay: registered worker **W-001** recognised; NO HELMET, VEST, GLOVES -> High alert



Phone camera, Warehouse (picture rotated upright): **same worker W-001** on a second camera; NO VEST



Phone camera: unregistered person gets a temporary visitor ID **V-001** (never stored)



Match / lighter flame held in the hand (24 Sep): **FIRE confirmed**, siren + spoken evacuation call

4.3 Autonomous response (what the laptop says)

Event	Announcement (Windows voice)
Registered worker violates PPE	'Attention worker W 0 0 1, Chinmaya. Please wear your helmet and safety vest now.' (every 20 s until compliant or acknowledged)
Unregistered person	'Attention unregistered person in the Welding bay. Please wear your gloves now.'
Fire / smoke confirmed	Siren + 'Fire alarm. Fire detected in the Welding Bay. Evacuate the Welding Bay now.'

5. Measured results

5.1 Inference speed (NVIDIA GeForce GTX 1650, 640 px, 20 real frames, measured 2026-10-12 08:15)

Model	Runtime	mean ms	p95 ms
person (yolov8n COCO)	PyTorch CUDA FP32	11.2	15.5
ppe main ppe_v2.pt (yolov8s)	PyTorch CUDA FP32	21.5	23.7
fire (fire_smoke_v2B.pt)	PyTorch CUDA FP32	9.2	14.2
person+keypoints (yolov8n-pose)	PyTorch CUDA FP32	11.3	17.1
ppe_aux ppe_v3B.pt (yolov8s-P2)	PyTorch CUDA FP32	28.0	29.6
shoe_cls (yolov8n-cls, 128, per foot crop)	PyTorch CUDA FP32	5.8	7.7

PyTorch on the GPU is 4-6x faster than ONNX on the CPU on this laptop, so the pipeline runs on the GPU. ONNX CPU numbers show what a GPU-less mini-PC would get. TensorRT / Jetson were not tested.

5.2 Several cameras at once (4-stream soak, 24 Sep, synthetic clips)

Camera	Zone risk	Target fps	Achieved fps
Welding bay	high	10	7.0
Assembly line	high	10	6.9
Warehouse	medium	10 -> 4 (load manager)	4.0
Warehouse 2	medium	10 -> 4 (load manager)	4.1

About 22 processed frames/s in total on one 4 GB GPU; lower-risk zones are slowed first, no stream stops.

5.3 Response time and other measurements

Measure	Value	How measured
Violation first seen -> alert (one camera)	median 2.8 s	event log, 24 Sep night shift
Violation first seen -> alert (webcam + phone)	median 4.3 s	event log, 10 Oct (two cameras share the GPU)
Edge -> dashboard transport	98-175 ms live; 0.32 s avg 10 Oct	captured_at vs received_at per alert
Alert size incl. snapshot	41.7 KB	329 alerts in the outbox
Single webcam processing rate	about 9.5 fps (on charger)	dashboard health; on battery it drops to about 2 fps
Live smoke false alarms rejected (replay)	26 / 30: all 25 older cases (faces, hair, flash); 4 new white-T-shirt cases from 10 Oct NOT yet	replayed from stored snapshots; rules were tuned on the older cases
Worker identity offline check	W-001 on 6 / 6 snapshots; 3 others -> visitor IDs	6 images, not a benchmark
Self-tests	51 of 52 pass	python -m edge.selftest; the failing one replays the 4 open white-T-shirt smoke cases

Accuracy of the models on our own footage (hard test set) is **not measured yet**; it needs the team's labelled data (planned). Model validation scores from training: PPE mAP50 0.872, fire/smoke mAP50 0.769 (public validation sets, not our cameras).

6. Robustness: failure-case gallery (real cases from our tests)

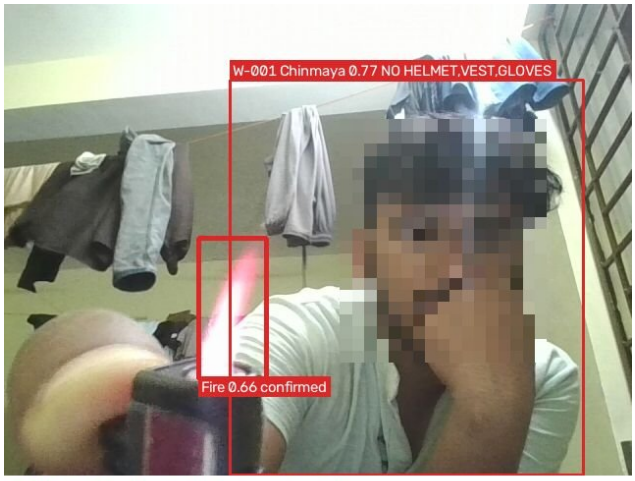
Each case shows the tricky input, what a naive detector does, and what NETRAX does now. The verdict under each image was computed by the current code (python -m tools.gallery).



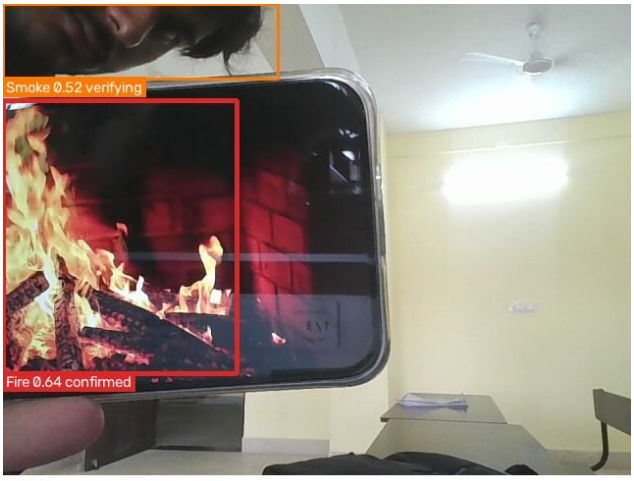
01. Worker's face and bare shoulders in a warm-lit room
Naive: The smoke model fires at conf 0.57: a Critical SMOKE alarm on a person.
Now: Colour check: 25% or more skin-coloured pixels means a face or arm, never smoke. Rejected.
Computed: REJECTED - skin 31% (a face / arm, not smoke)



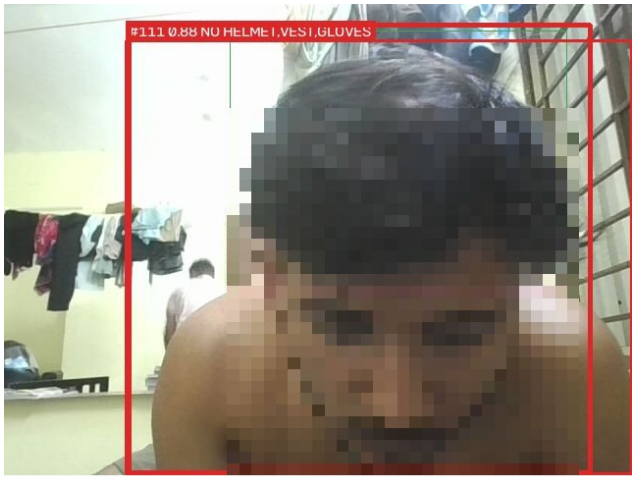
03. Phone flash pointed at the camera
Naive: The bright haze is read as smoke, giving a Critical alarm.
Now: 25% or more blown-out pixels means glare from a flash or lamp. Rejected.
Computed: REJECTED - glare (phone flash / lamp)



04. Match / lighter flame held in the hand, close to the camera
Naive: Old rule: fire on a person = orange clothing, so it was suppressed and no alarm sounded.
Now: A flame glows (white-hot core) and flickers; clothing does neither. CONFIRMED, siren on.
Computed: PASSES colour check - flame colour 14%, hot core 35%



05. Phone playing a fireplace video
Naive: Fire alarm.
Now: KNOWN LIMITATION: still confirmed as fire. A camera alone cannot tell a screen from a flame. Fix planned: an IR flame sensor in the hardware phase.
Computed: PASSES colour check - flame colour 19%, hot core 2%



06. Seated close-up, legs out of the frame

Naive: 'No safety shoes' alarm: the feet are not visible, so no shoes are detected.

Now: Feet not in view means 'unverifiable', never 'missing'. No alert.

Computed: boots -> unverifiable



07. Registered worker next to an unregistered visitor

Naive: The tracker gives a new number every time someone re-enters, so the same person gets many alerts.

Now: Face match against registration photos: the worker is always W-001, the visitor gets a temporary V-id. The speaker calls the worker by ID.

Computed: offline identity run: W-001 on 6/6 snapshots of the registered worker (docs/model_card.md)

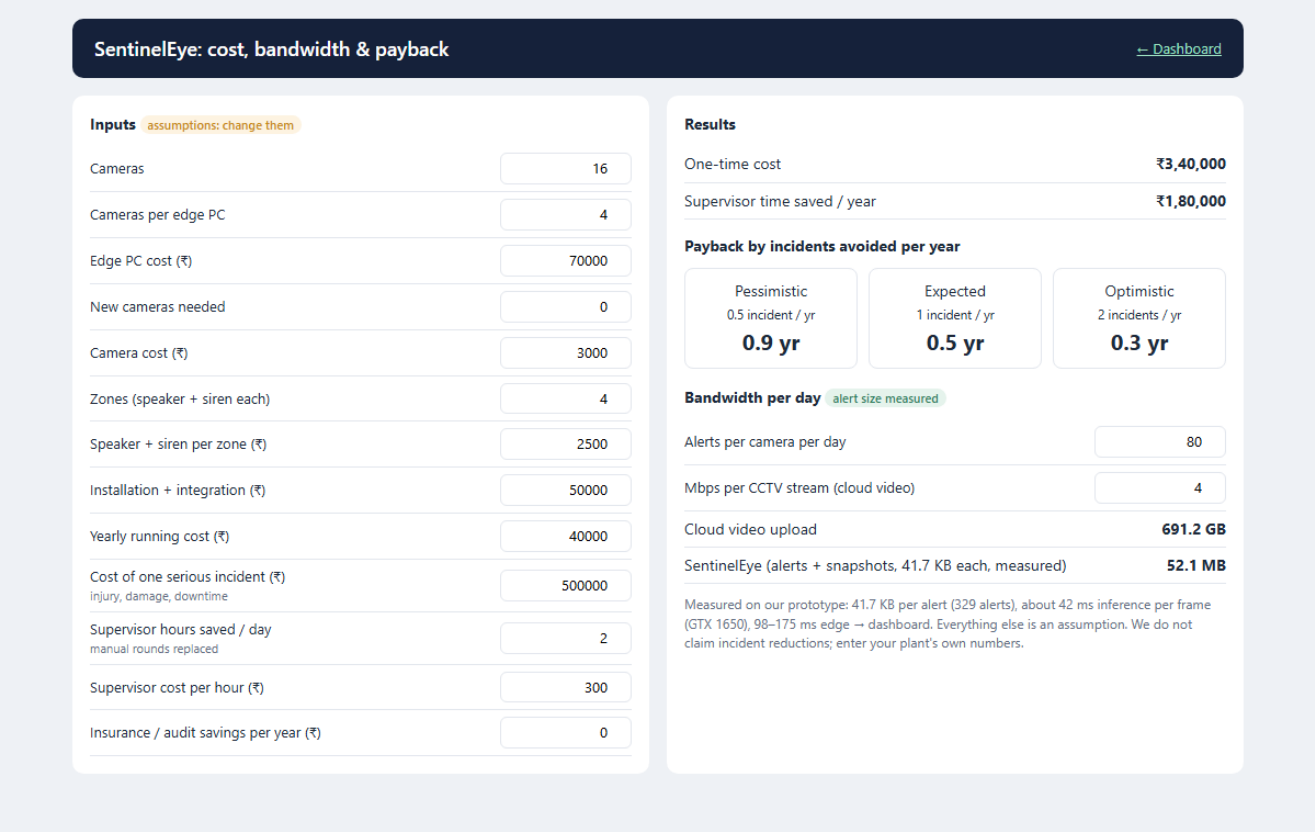
7. Problems found in live tests and how they were fixed

Problem seen in live tests	Cause	Fix	Status
PPE alert took about 4 s	10/15 frames + 3 s grace; outbox polled every 0.3 s	7/10 frames + 1.5 s; instant outbox; CUDA warm-up	**Fixed**
Speaker called only the first worker	TTS library spoke once then went silent	Windows SAPI voice directly; queue per announcement	**Fixed**
Speaker stopped after 3 calls	Max 2 repeats	Repeat every 20 s until compliant / acknowledged	**Fixed**
No new alert when the same worker violated again	60 s cooldown meant for tracker ID switches	Re-arm: same worker, 5 s after fixing -> new alert	**Fixed**
New ID every time a person re-entered	Tracker numbers restart	Face registration -> one stable ID per person	**Fixed**
Match / fire close to camera ignored	'Fire on a person = orange vest' rule	Flame must glow + flicker; white-hot core counted	**Fixed**
Agarbatti smoke never confirmed	Thin plume kept starting new regions; high threshold	Region matching tolerates drift; lower threshold; every frame when busy	**Fixed (needs live re-check)**
Smoke alarms on faces / hair / flash	Model confuses them with smoke	Skin, dark-hair and glare checks	**Fixed (25/25 old cases)**
Fire core also reported as smoke	White flame core looks like smoke	Smoke inside a fire box = part of the fire	**Fixed**
False 'no safety shoes' in close-ups	Feet out of frame	Full-body check; shoes paused anyway	**Fixed**
Phone picture sideways	Phone held upright	rotate: 90 option per camera	**Fixed**
Unreachable phone slowed the webcam	Stream open hung > 60 s	1.5 s reachability check first	**Fixed**
Smoke false alarms on a person in a white T-shirt (10 Oct)	Bright, low-colour shirt passes smoke checks	To do: stronger on-person rule / own data	**Open**
Black gloves reported missing (10 Oct)	Public glove data has few black gloves	Own glove data + retraining	**Open**

Every fix has an automated regression test in edge/selftest.py.

8. Deployment, bandwidth and cost

Item	Value	Source
Cloud video, 16 cameras x 4 Mbps	about 690 GB per day upload	assumption (2-6 Mbps per camera is typical)
NETRAX, 16 cameras x 80 alerts x 41.7 KB	about 53 MB per day	alert size measured
Edge PC with GPU, per ~4 cameras	Rs 70,000 (example)	assumption
Payback	one-time cost / (yearly benefit - running cost)	formula; calculator on the dashboard



ROI calculator (dashboard -> footer link). All rupee values are placeholders to change; we do not claim incident reductions.

9. Known limitations (honest list)

Limitation	Plan
A phone screen playing a fire video is detected as fire	IR flame sensor in the hardware phase (camera AND sensor)
Smoke false alarms on a white T-shirt; black gloves missed	Own data + retraining; stronger on-person smoke rule
Safety shoes switched off	Train on our own shoe photos, then re-enable
Face-based worker ID; judging checklist prefers badge/RFID	Consent-based and switchable; RFID badge in hardware phase
Accuracy on our own footage not measured yet	Hard test set from the team's data (tools ready)
Slower with several cameras / on battery	Load manager; run on charger; ONNX/TensorRT not used on this GPU
Notifications are MOCK unless Telegram is configured	Configure a bot token for real messages

10. What comes next

#	Work	Needs
1	Record own data (shoes, black gloves, white shirts, fire/smoke, negatives), label, build dataset, evaluate, retrain v3	Team recordings (tools + guide ready)
2	Record demo clips for the 3rd camera (fire video for Assembly line)	Team recording
3	Fix open issues from 10 Oct (white-shirt smoke, black gloves)	Rules now, data for gloves
4	Slides, backup demo video, rehearsal (docs/demo_script.md)	-
5	Hardware phase: IR flame + MQ-2 smoke sensors, RFID, ESP32 siren (docs/hardware_plan.md)	Hardware parts